MBA – Market Basket Analysis

Market Basket Analysis (MBA) is a technique used in retail to find relationships between products that are frequently bought together.

For Eg:

"If a customer buys **Product A**, what else are they likely to buy?"

This is done using **association rule mining** on **POS (Point-of-Sale)** data.

Use Cases of MBA :

Use Case	Description
• **Cross-selling**	• Suggest related products (e.g., Buy phone → Recommend case + charger)
• **Bundling Offers**	• Combine high-frequency items into discount combos (e.g., Milk + Bread)
• **Checkout Recommendations**	• Dynamic product suggestions at billing counter or online cart
• **Shelf Placement**	• Place co-purchased items closer to each other in aisles
• **Marketing Campaigns**	• Targeted promotions based on item associations

Tools You Can Use:

Layer	Tool/Library
Data Handling -	Python, pandas, Excel
Algorithm -	mlxtend, apriori, FP-Growth
Visualization -	Power BI
Dashboard -	Power BI
Storage -	(MySQL / PostgreSQL)

Libraries in python:

1. Pandas – For data wrangling

2. mlxtend – For Apriori & rule mining

3. TransactionEncoder – Converts item lists to binary matrix

4. association_rules() – To extract rules from itemsets.

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Algorithms:

Algorithm	Description
Apriori	Classic, simple to understand, slower on big data
FP-Growth	Faster than Apriori, suitable for large datasets
ECLAT	Less common but useful for sparse data

How It Works – Step-by-Step:

1. Collect POS Transactions (e.g., CSV or DB)

2. Preprocess: Clean & structure data

3. Convert item lists into binary matrix (1 = present, 0 = absent)

4. Apply Apriori or FP-Growth to find frequent itemsets

5. Generate association rules (A → B)

6. Filter rules by:

- **Support ≥ 0.3**
- **Confidence ≥ 0.6**
- **Lift > 1**

7. Visualize results in Power BI

8. Apply findings to business (bundles, layout, marketing)

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Work Flow Diagram:

[POS Data]

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[Preprocessing in Python (pandas)]

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[Item Encoding (TransactionEncoder)]

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[Apriori Algorithm (mlxtend)]

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[Association Rule Mining]

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[Filtered Rules (by lift/confidence)]

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[Visualization (Power BI / Tableau)]

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[Business Strategy: Bundle / Cross-Sell / Shelf Layout]

Example Rule Output:

Rule	Support	Confidence	Lift
{Milk} → {Bread}	0.60	0.85	1.2
{Diaper} → {Beer}	0.50	0.66	1.4

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How It Helps a Supermarket

Area	Benefit
Sales Boost	Increases average cart value via bundles
Checkout	Dynamic cross-sell recommendations
Shelf Planning	Better aisle organization to drive co-purchases
Promotions	Run combo deals backed by data
Inventory Planning	Predict commonly sold-together items for stock management

Key Metrics: Support, Confidence, and Lift

These three metrics are essential in evaluating the **usefulness and reliability** of association rules.

Metric	Definition	Use in Business
Support	Proportion of transactions that include both A and B	Helps identify how **frequently** the itemset occurs in the data
Confidence	Likelihood that a customer who buys A also buys B	Measures the **reliability** of the rule (A → B)
Lift	How much more likely B is bought when A is bought, compared to B being bought independently	If **Lift > 1**, A and B have a **positive correlation**

Example:

Rule: `{Milk} → {Bread}`

- **Support** = 0.60 → 60% of transactions contain both Milk and Bread
- **Confidence** = 0.85 → 85% of the time when Milk is bought, Bread is also bought
- **Lift** = 1.2 → Milk buyers are 1.2x more likely to buy Bread compared to a random shopper

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